

Supplementary Information:
Logic-V: Reinforcement Alignment and Selective
Multi-Expert Verification for
Resource-Constrained Multimodal Reasoning

Zexi Chen¹, Zhendong Li^{1*}, Shuai Li¹, Hao Liu¹, Yiqiang Wu¹

¹ School of Information Engineering, Ningxia University, Yinchuan,
750021, China .

*Corresponding author(s). E-mail(s): lizhendong@nxu.edu.cn;

Online Resource 1. This document accompanies the manuscript submitted to *Applied Intelligence*. It provides extended generalization results, cross-benchmark role analysis, evidence-extraction details, training dynamics, qualitative examples, deployment measurements, training hyperparameters, prompt templates, and the MedQA transfer probe. Tables, figures, equations, and sections are numbered independently with the prefix “S.” The terminology and experimental configurations follow the main paper: the training-time blind-review judge is a frozen Qwen2.5-7B-Instruct model, whereas the inference-time external verifier is a frozen Qwen2.5-32B-Instruct-AWQ model in INT4 form.

Section S1 reports out-of-domain video results, cross-benchmark role collaboration, and external-verifier activation rates. Section S2 specifies the Task-Adaptive Perception Protocol (TAPP). Section S3 presents GRPO training dynamics. Section S4 compares evidence-extraction modes. Section S5 provides qualitative success and failure cases. Section S6 reports stage-wise memory and latency measurements. Section S7 lists reproducibility settings. Section S8 documents the ordered EgoSchema CAHV configuration used in the fixed-perception and MedQA analyses. Section S9 provides the virtual-expert and external verifier prompts together with the MedQA transfer analysis.

S1 Cross-Domain Generalization Results

S1.1 MVBench Generalization

Table S1: Generalization on MVBench [Li et al \(2024\)](#) (420 questions from seven task subsets). All decoupled language-model baselines consume identical textual evidence logs extracted by Qwen2.5-VL-7B-Instruct [Qwen Team \(2025\)](#). AVF reaches 45.0%, exceeding the corresponding 3B-SFT baseline by 5.9 percentage points. The direct-video Qwen2.5-VL-7B result is included as a reference and is not a controlled reasoning-stage comparison

Method	Size	Input	Avg. (%)
<i>Direct-video reference</i>			
Qwen2.5-VL-7B-Instruct Qwen Team (2025)	7 B	Video	54.7
<i>Decoupled systems with identical evidence logs</i>			
Qwen2.5-7B zero-shot Yang et al (2024)	7 B	Text log	46.7
Qwen2.5-3B zero-shot Yang et al (2024)	3 B	Text log	26.4
3 B-SFT	3 B	Text log	39.1
AVF (Ours)	3 B	Text log	45.0
Improvement over 3 B-SFT (Δ)			+5.9

S1.2 Video-MME Long-Form Generalization

Table S2: Generalization on the long-video subset of Video-MME [Fu et al \(2024\)](#) (900 questions from videos of at least 30 minutes). All decoupled language-model baselines consume the same Mode III textual evidence logs. AVF reaches 37.1%, exceeding the corresponding 3 B-SFT baseline by 11.9 percentage points. Direct-video systems are reported as references rather than controlled reasoning-stage baselines

Method	Size	Input	Avg. (%)
<i>Direct-video references</i>			
Video-LLaMA2-7B-16F Cheng et al (2024)	7 B	Video	45.4
Qwen2.5-VL-7B-Instruct Qwen Team (2025)	7 B	Video	51.5
<i>Decoupled systems with identical evidence logs</i>			
Qwen2.5-7B zero-shot Yang et al (2024)	7 B	Text log	39.7
Qwen2.5-3B zero-shot Yang et al (2024)	3 B	Text log	30.4
3 B-SFT	3 B	Text log	25.2
AVF (Ours)	3 B	Text log	37.1
Improvement over 3 B-SFT (Δ)			+11.9

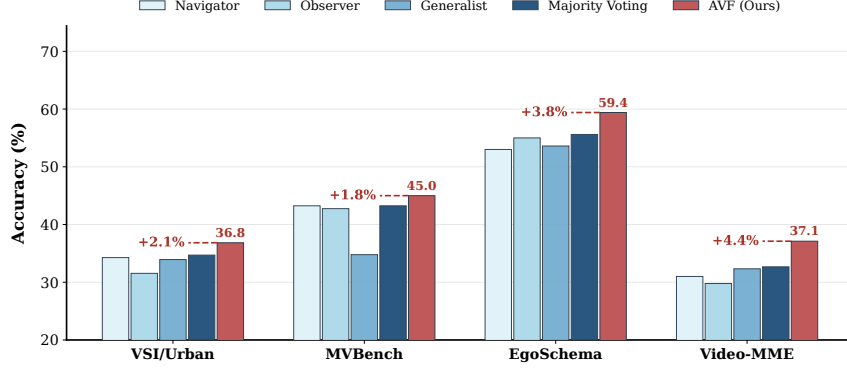


Fig. S1: Role-conditioned collaboration across four video benchmarks. The 30-example calibration selects Observer as the preferred specialist for VSI/Urban, MVBench, and Video-MME, and Navigator for EgoSchema. Full-split evaluation shows that the calibration-selected specialist is not always the strongest individual role. Nevertheless, AVF with CAHV exceeds naive majority voting by 1.8–4.4 percentage points across the four evaluated settings. The result indicates practical tolerance to imperfect small-set role selection without implying calibration invariance

S1.3 Multi-Expert Collaboration Across Benchmarks

S1.4 External-Verification Rates

Table S3 reports the fraction of queries routed to the frozen 32 B-INT4 external verifier. The rates are reported separately because CAHV uses benchmark-specific calibration rules fixed before test evaluation.

Table S3: Dataset-specific external-verification rates. Non-verifier resolution includes fast-path consensus, specialist override, and Generalist fall-back

Benchmark	External-verifier rate	Non-verifier resolution
VSI/Urban	22.5%	77.5%
EgoSchema	14.0%	86.0%

Table S4: Detailed EgoSchema routing decomposition. Fast-path consensus accounts for 76% of queries, while the complete non-verifier total is 86% because specialist override and Generalist fallback also avoid the 32 B verifier

Route	Share of queries	Uses 32 B verifier?
Fast-path consensus	76%	No
Specialist override	6%	No
Generalist fallback	4%	No
Verifier-escalated disagreement	14%	Yes
Non-verifier total	86%	No

S2 Task-Adaptive Perception Protocol

This section specifies the three deterministic evidence-extraction modes summarized in the main paper. Mode assignments are fixed from benchmark characteristics before evaluation and are not selected using test labels.

Mode I: Global-Glimpse

Mode I forms an eight-frame pooled view for short clips of approximately 30 seconds. The fixed frame budget prioritizes broad semantic coverage over dense temporal transitions. In this work, Mode I is used for MVBench.

Mode II: Sequential Differential Scan

Mode II extracts pairwise differential states with stride $w = 2$ from a 32-frame sequence for motion-sensitive content. It has two evaluated variants. The query-blind VSI/Urban variant withholds the question from the perception model so that evidence extraction is independent of the answer query. The query-aware EgoSchema variant supplies the question to the 72 B-INT4 perception model and uses a 24-frame budget to control the volume of long-form evidence.

Mode III: Discrete Anchors

Mode III constructs a sparse, query-aware anchor set $\mathcal{K} \subset \{1, \dots, T\}$ for videos of at least 30 minutes. The sparse anchors limit evidence volume while retaining temporally distributed observations. Mode III is used for the long-video subset of Video-MME.

Benchmark-to-mode assignment

VSI-Bench and UrbanVideo-Bench use Mode II query-blind extraction. MVBench uses Mode I. EgoSchema uses Mode II query-aware extraction with a 72 B-INT4 perception backbone. Video-MME uses Mode III query-aware discrete anchors. Complete-system comparisons among the evaluated modes are reported in Table S5.

S3 Full GRPO Training Dynamics

Figure S2 supplements the main-paper reward comparison with accuracy, format, lexical stability, generation length, and end-to-end ablation curves from the reported training run.

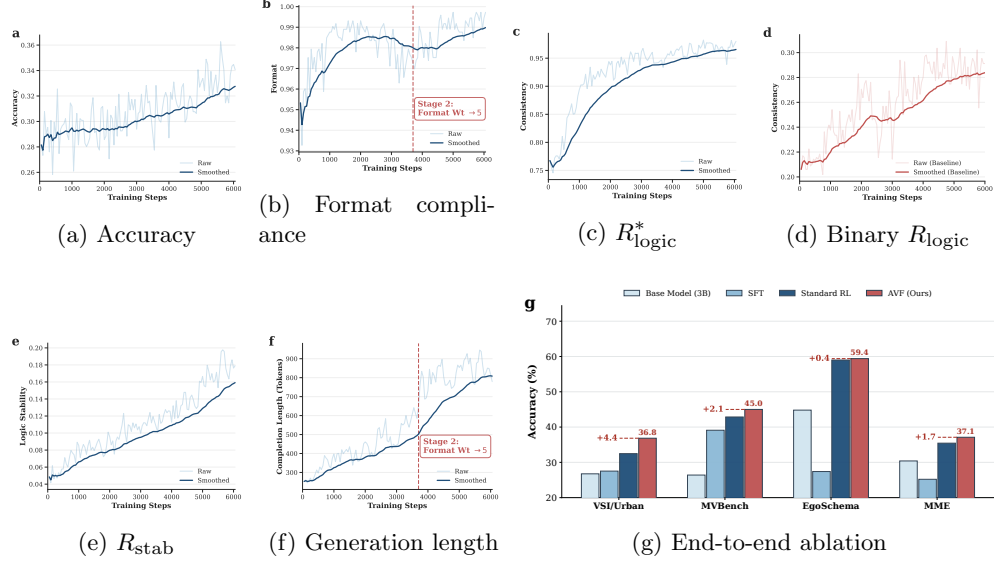


Fig. S2: GRPO training dynamics and end-to-end ablation. Panels (a)–(d) show accuracy, format compliance, and the two trajectory-to-decision consistency rewards. The recovery in panel (b) follows the increase $\lambda_{\text{fmt}} : 1 \rightarrow 5$ at step 3700. Panels (e)–(g) show lexical trajectory stability, generation length, and the cumulative system ablation. Panels (c) and (d) are shown as raw and smoothed trajectories from the same training seed; the entropy-aware curve appears visually smoother, but the figure is not a multi-seed variance estimate

S4 Evidence-Extraction Mode Ablation

The results support the fixed benchmark-to-mode assignments used in the reported experiments. They do not establish that a mode is universally optimal outside the evaluated benchmarks or extraction budgets.

Table S5: Complete-system accuracy under the evaluated evidence-extraction modes on MVBench Li et al (2024), EgoSchema Mangalam et al (2023), and the long-video subset of Video-MME Fu et al (2024). Bold values mark the highest observed accuracy among the modes evaluated for each benchmark. Dashes denote mode-benchmark combinations not evaluated by design

Temporal setting	Benchmark	Mode I	Mode II	Mode III
<i>Short horizon (approximately 30 s)</i>				
Short video	MVBench	45.0	37.0	—
<i>Egocentric causal reasoning</i>				
Long egocentric video	EgoSchema	48.2	59.4	—
<i>Long form (at least 30 min)</i>				
Long video	Video-MME	—	33.2	37.1

S5 Qualitative Reasoning Traces



Fig. S3: Qualitative AVF reasoning traces. (a) In the success case, the external verifier rejects a constraint-blind candidate after checking it against the action log and returns the evidence-supported alternative. (b) In the failure case, strict evidence enforcement produces an overly literal interpretation of a counting question that depends on an implicit human convention. The contrast illustrates both the value and the limitation of hostile evidence verification and motivates future work on task-adaptive evidence permissiveness

S6 Computational Profile and Deployment Costs

Table S6 reports stage-wise memory and latency on EgoSchema. The memory values are not additive because perception, compact reasoning, and external verification are executed sequentially. The end-to-end single-GPU requirement is therefore the maximum measured stage peak.

Table S6: Stage-wise computational profile on EgoSchema [Mangalam et al \(2023\)](#). The Embodied-R row is a reported reference reproduced from Embodied-R [Zhao et al \(2025\)](#). The AVF stages are measured under sequential loading on one 48 GB GPU. Peak-memory values are not additive

Stage or route	Configuration	Loading policy	Peak VRAM (GB)	Latency (s/query)
<i>Reported reference</i>				
Embodied-R Zhao et al (2025)	72 B VLM (32 frames, FP16) + 3 B LM	reported pipeline	~145.0	243.68
<i>AVF with sequential stage loading</i>				
Evidence extraction	72 B-INT4 perception, 24 frames	perception-only stage	42.5	348.39
Reasoning without verifier (86%)	3 B shared virtual experts	perception offloaded; verifier absent	6.8	155.03
Verifier-escalated reasoning (14%)	3 B experts + 32 B-INT4 verifier	perception offloaded; verifier loaded on demand	38.5	165.06
End-to-end stage-wise profile	weighted by 14% verifier activation	sequential execution	$\max(42.5, 38.5) = 42.5$	504.82

The expected latency is computed from the one-pass perception cost, the shared compact-policy reasoning cost, and the incremental verifier overhead on the 14% escalated subset:

$$\begin{aligned}
 T_{\text{dyn}} &= T_{\text{perception}} + T_{\text{reasoning}} + p_{\text{ver}} (T_{\text{verifier route}} - T_{\text{reasoning}}) \\
 &= 348.39 + 155.03 + 0.14(165.06 - 155.03) \\
 &= 504.82 \text{ s/query.}
 \end{aligned} \tag{S1}$$

The measured 42.5 GB end-to-end peak occurs during 72 B-INT4 evidence extraction; subsequent AVF reasoning and verification peak at 38.5 GB. For VSI/Urban, the corresponding measured stage-wise peak is 38.5 GB with a 22.5% verifier rate. These measurements support a peak-memory and selective-activation claim, not a latency, total-computation, or energy advantage. In particular, the reported EgoSchema latency is higher than the Embodied-R reference because AVF samples 15 compact-policy trajectories before routing.

S7 Extended Training and Reward Settings

Table S7: Training, reward, and implementation settings used for the reported Logic-V checkpoint

Setting	Value
<i>Policy optimization</i>	
Trainable policy	Qwen2.5-3B-Instruct Yang et al (2024)
Learning rate	5×10^{-6}
LR scheduler	Cosine with 5% warmup
Batch size per policy GPU	4 prompts
Gradient accumulation	4 steps
Maximum input length	8,192 tokens
Maximum generation length	2,048 tokens
GRPO group size	$K = 4$ trajectories per prompt
KL coefficient	$\beta = 0.005$
Sampling temperature	0.7
LoRA rank and scale	$r = 32, \alpha = 64$
LoRA target modules	All linear layers
Training epochs	12
Total optimization steps	6,000
Training seed	3407
Training hardware	Two 48 GB NVIDIA Ampere vGPU instances
<i>Blind-review judge and reward implementation</i>	
Frozen judge	Qwen2.5-7B-Instruct Yang et al (2024)
Judge input	Question q and generated trajectory z only
Hidden from judge	Student decision y , ground truth y_{gt} , and original evidence log s
Judge decoding	Deterministic generation (<code>do_sample=False</code>)
Answer-label vocabulary	$\{A, \dots, H\}$
Maximum label entropy	$\log_2 8$
Stability similarity	Lexical-set Jaccard similarity
Similarity threshold	$\tau_{sim} = 0.3$ with strict $>$ comparison
Minimum trajectory length	50 characters
Target trajectory length	1,000 characters
<i>Reward configurations</i>	
G-I: baseline rewards	$(R_{acc}, R_{fmt}, R_{logic})$, weights (7, 1, 1)
G-II: add stability	$(R_{acc}, R_{fmt}, R_{logic}, R_{stab})$, weights (7, 1, 1, 1)
G-III Stage 1	$(R_{acc}, R_{fmt}, R_{logic}^*, R_{stab})$, weights (7, 1, 1, 1)
G-III Stage 2	$\lambda_{fmt} : 1 \rightarrow 5$ at step 3700
<i>Inference-time verification</i>	
Shared role policy	Qwen2.5-3B-Instruct Yang et al (2024)
Role samples	$N = 5$ per role; 15 compact-policy trajectories in total
Frozen external verifier	Qwen2.5-32B-Instruct-AWQ, INT4 Yang et al (2024)

S8 CAHV Ordered Routing Configuration

This section documents the ordered EgoSchema CAHV configuration used in the fixed-perception analysis and retained unchanged in the MedQA transfer probe. It follows the released evaluation implementation. Let y_g , y_n , and y_o denote the Generalist, Navigator, and Observer role-level decisions, and let g , n , and o denote their respective support counts among five sampled trajectories. Let v_1 and v_2 denote the largest and second-largest pooled vote counts across all valid trajectories, with $\Delta_v = v_1 - v_2$. Rules are evaluated in the order shown in Table S8; the first satisfied rule returns immediately.

Table S8: Ordered local-decision rules in the released EgoSchema CAHV configuration. “Top-tier” means tied for the largest pooled vote count. A dissent condition is ignored when the corresponding role yields no valid answer

Order	Condition	Returned decision
1	$y_n = y_o$ and $n + o > 9$	y_n
2	$n = 5$, $g \leq 3$, and $o \leq 3$	y_n
3	$g \leq 2$, $n \geq 4$, $y_n \neq y_g$, and $o < 4$	y_n
4	$y_g = y_n$, $g + n > 6$, and $o < 3$	y_g
5	$y_g = y_o$, $g + o > 7$, and $n \leq 2$	y_g
6	$g = 5$, y_g is the pooled leader, and $\Delta_v > 1$	y_g
7	$g = 4$, with neither Navigator dissent ($y_n \neq y_g$, $n \geq 3$) nor Observer dissent ($y_o \neq y_g$, $o \geq 4$)	y_g
8	$o \geq 4$, y_o is top-tier, $y_o \neq y_g$, $g \leq 2$, and no Navigator dissent ($y_n \neq y_o$, $n \geq 3$)	y_o
9	$y_g = y_n$, $g + n \geq 7$, and $o \leq 3$	y_g
10a	$\Delta_v = 1$ and the pooled leader is y_n	y_n
10b	$\Delta_v = 1$, the pooled leader is y_o , and $n < 3$	y_o
10c	$\Delta_v = 1$, the pooled leader is y_g , and $n < 3$	y_g

If no local rule is satisfied, CAHV constructs a two-candidate verification problem using the ordered cases in Table S9. The representative rationale for a role is the longest valid generated rationale supporting that role’s majority decision.

Table S9: Ordered challenger construction and external-verification behavior in the released EgoSchema CAHV configuration

Order	Condition	Candidate X	Candidate Y
A	$g \leq 2$, $y_n \neq y_o$, $n \geq 3$, and $o \geq 3$	Navigator decision	Observer decision
B	$y_n = y_o \neq y_g$	Generalist decision	Shared specialist decision
C	$y_n \neq y_g$ and $n \geq 3$ when $g \geq 4$, otherwise $n \geq 2$	Generalist decision	Navigator decision
D	$y_o \neq y_g$ and $o \geq 4$ when $g \geq 3$, otherwise $o \geq 3$	Generalist decision	Observer decision

When none of the challenger cases applies, CAHV returns the Generalist decision without loading the external verifier. Otherwise, the verifier receives the question, candidate descriptions, the two representative supporting rationales, and a budgeted evidence context. The implementation reserves 1,024 tokens for the verifier response. If the evidence exceeds the remaining context budget, it retains the first 800 evidence tokens and the final portion that fits the residual budget, separated by an explicit skipped-content marker. A valid verifier decision must correspond to one of the candidate answer labels; an invalid or failed verifier response falls back to the Generalist decision. Because the rule cascade returns after its first match and constructs only one candidate pair, each query enters external verification at most once.

S9 Virtual Expert and External-Verifier Prompt Templates

The three virtual experts—Generalist, Navigator, and Observer—share the same Qwen2.5-3B-Instruct weights Yang et al (2024); only the role prompt changes. The roles therefore require no separately trained or simultaneously resident expert networks, although AVF still performs five decoding passes per role. The frozen external verifier runs Qwen2.5-32B-Instruct-AWQ in INT4 form Yang et al (2024) and is invoked only after CAHV constructs a valid challenger. It receives the question, a

fixed-budget evidence context, candidate-answer descriptions, and one representative supporting rationale for each candidate. When the evidence exceeds the verifier context budget, the implementation retains its beginning and end.

For readability, the prompt templates below omit Markdown heading markers and emphasis delimiters and apply minor typographic and linguistic normalization. These presentation changes do not alter the intended role specialization or output constraints. The exact machine-readable templates used in the experiments are provided in the released implementation.

S9.1 Generalist (E_{gen})

Generalist prompt

You are an expert video analyst with strong logical reasoning skills. Your task is to analyze the provided Video Log and answer the multiple-choice question.

Critical reasoning rules

1. **Specificity over generality.** If Option A is “Cleaning the kitchen” (broad) and Option B is “Washing dishes” (specific), and the video only shows washing dishes, choose Option B.
2. **Cycle completion for manual tasks.** In cyclical manual labor, such as making bricks or cooking, if the log shows the beginning of a cycle but ends before completion, assume that the visible cycle completes. Prefer an option describing the complete cycle, including cleanup or reset, over one that terminates prematurely, provided that the beginning of the cycle is visible.
3. **Evidence grounding.** Do not introduce objects or events that are absent from the video log.

Output requirements

1. First, provide step-by-step reasoning inside `<think>` and `</think>` tags. Check explicitly for specificity and cyclical patterns.
2. Then provide the final answer inside `<answer>` and `</answer>` tags. The answer must contain one option letter only.

Example

Question: What is the person doing?

Options: A. Making bricks and cleaning the mold. B. Making bricks.

Video Log: Frame 0: Person smooths sand. Frame 10: Person fills the mold. Frame 20: Person drops the brick. End of log.

Expected response:

`<think>` The log depicts a cyclical task. Frame 0 shows preparation or cleaning through smoothing the sand. Option A includes the cleanup component implied by the visible cycle, whereas Option B describes only part of the process. `</think>`

`<answer>` A `</answer>`

Evaluation input

Video Log: [Video Log]

Question: [Question]
Options: [Options]

S9.2 Navigator (E_{nav})

Navigator prompt

You are a Spatial Navigation Specialist. Your task is to analyze the agent’s movement trajectory and spatial orientation over time.

Evaluation input

Video Log: [Video Log]
Question: [Question]
Options: [Options]

Reasoning instructions

1. **Trajectory mapping.** Identify the sequence of move, turn, and rotation actions. Determine where the agent starts and where it ends.
2. **Spatial consistency.** When the question concerns a relative position such as behind or left, trace the agent’s orientation changes before determining the answer.
3. **Logical elimination.** Discard options that contradict the physical path recorded in the evidence log.
4. **Physical action over broad labels.** Prioritize the observed movement. For example, when the person moves plates, prefer “moving plates” over a broad description such as “cleaning the house.”

Output requirements

Provide step-by-step trajectory analysis inside `<think>` and `</think>` tags, followed by one option letter inside `<answer>` and `</answer>` tags.

S9.3 Observer (E_{obs})

Observer prompt

You are a Visual Evidence Analyst. Your task is to verify the existence, state, and interaction of objects in the video log.

Evaluation input

Video Log: [Video Log]
Question: [Question]
Options: [Options]

Reasoning instructions

1. **Fact checking.** Inspect the log for objects and actions mentioned in the candidate options. An option requiring an object that is never observed should receive reduced support.
2. **Implicit actions.** When the log records a result state, such as a clean mold, infer the most direct preceding action when that inference is supported by the visible evidence.

3. **Evidence-based elimination.** Eliminate options that require objects or interactions unsupported by the evidence log.

Output requirements

Provide evidence-based reasoning inside `<think>` and `</think>` tags, followed by one option letter inside `<answer>` and `</answer>` tags.

S9.4 External Verifier (V_{32B}): Hostile Verification Authority

External-verifier prompt

You are a Forensic Video Logic Analyst. Your goal is to identify the most defensible answer by analyzing the underlying physical actions rather than relying on unsupported narrative detail.

Core principle: physical evidence over narrative embellishment

The video log may be incomplete. Candidate rationales may introduce specific game titles, intentions, or adjectives that are not directly supported by the evidence. Ignore unsupported descriptive detail and focus on observable hand-object interactions and physical events.

Evidence rules

1. **Flavor-trap rule.** Penalize candidate descriptions that introduce unsupported attributes, intentions, or domain-specific labels.
For example, when one candidate describes “playing a strategic board game involving resource management” and another describes “placing cards on a table,” prefer the latter when only the physical placement of flat objects is visible.
2. **Interaction rule.** Prefer the candidate whose required object shape and physical interaction most closely match the evidence log.
For example, when the log records a hand moving a flat object onto a table, a card-related description is better supported than one requiring unobserved pieces or dice.
3. **Clear-absence rule.** Reject a candidate because of missing evidence only when it requires an object or action that is clearly absent. Do not reject a semantically compatible candidate merely because the evidence log uses a less specific noun.

Verifier input

Question

[Question]

Budgeted evidence context

[Video Log]

Candidate X

Description: [Candidate X description]

Supporting rationale: [Candidate X reasoning]

Candidate Y

Description: [Candidate Y description]

Supporting rationale: [Candidate Y reasoning]

Judgment procedure

1. **Remove unsupported detail.** Identify descriptive material that is not grounded in the evidence.

2. **Match physical mechanics.** Compare the observed movements, object states, and interactions required by the two candidates.
3. **Apply the simpler supported explanation.** When one candidate requires invisible assumptions and the other directly describes the observed evidence, prefer the latter.
4. **Return the verdict.** Choose the candidate that is most consistent with the available physical evidence.

Output requirements

Provide the comparison inside `<think>` and `</think>` tags. Return exactly X or Y inside `<answer>` and `</answer>` tags.

S9.5 Medical Domain Instantiation for the MedQA Pilot

For the 300-question MedQA-USMLE pilot [Jin et al \(2021\)](#), only the role descriptions are replaced by medical-domain analogues. The EgoSchema-calibrated CAHV thresholds and the Navigator specialist preference ($E_{\text{spec}} \leftarrow E_{\text{nav}}$) are retained unchanged. No medical examples are used for calibration or parameter updates. This experiment is therefore a structural cross-domain transfer probe rather than a clinical-performance claim.

Table S10: MedQA-USMLE structural transfer pilot on 300 questions. The Logic-V checkpoint was trained with seed 3407. No medical-domain calibration is used; the EgoSchema CAHV configuration is retained unchanged. The table reports system-level accuracy, while Figure S4 compares AVF and majority voting on the same role trajectories by routing path

Method	Reasoner	Medical calibration	Acc. (%)
Qwen2.5-3B zero-shot Yang et al (2024)	3 B	None	58.0
Qwen2.5-7B zero-shot Yang et al (2024)	7 B	None	69.0
Logic-V (single-pass)	3 B	None	56.6
AVF (Full)	3 B	None	63.7

S9.6 Medical-Domain Role and Verifier Prompts

The following templates instantiate the video-domain roles in the MedQA-USMLE pilot. They use the same presentation format as the video-domain prompts. Markdown markers are omitted and minor typographic normalization is applied for readability. The exact machine-readable templates used in evaluation are provided in the released implementation.

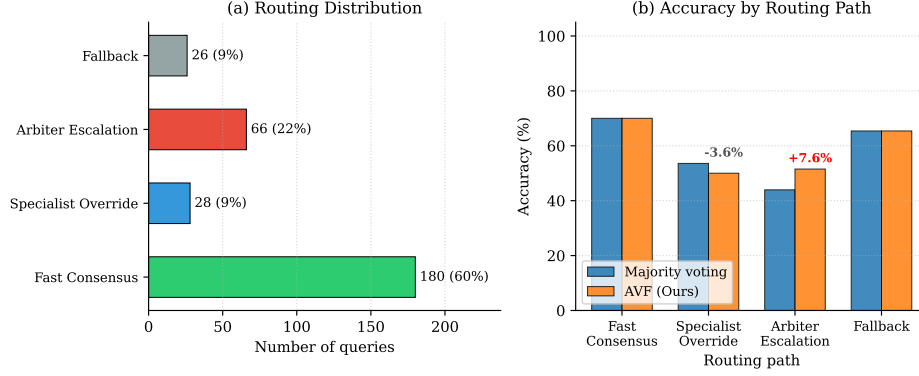


Fig. S4: Routing-path decomposition of the MedQA-USMLE pilot. (a) CAHV routes 60% of questions through fast-path consensus, 9% through specialist override, 22% through external verification, and 9% through Generalist fallback. (b) AVF is compared with majority voting over the same 15 role trajectories. Fast-path and fallback accuracy are unchanged. On the verifier-escalated subset, AVF reaches 51.5% versus 43.9% for majority voting, a gain of 7.6 percentage points. On the specialist-override subset, AVF reaches 50.0% versus 53.6%; this 3.6-point difference is reported descriptively without a statistical significance claim. Because both methods use the same compact-policy samples, the routed comparison isolates the decision rule applied after role generation

Internist (E_{int} , replacing Generalist)

Internist prompt

You are an experienced internist analyzing a clinical case. Your task is to answer the multiple-choice question by reasoning over the patient’s case description.

Critical reasoning rules

1. **Holistic assessment.** Consider the complete clinical picture rather than isolated findings.
2. **Specificity over generality.** Prefer the diagnosis that explains the largest number of findings simultaneously.
3. **Evidence-based reasoning.** Do not introduce conditions, symptoms, or findings that are not present in the case description.
4. **Differential reasoning.** Consider competing hypotheses and determine which alternatives can be supported or excluded by the stated evidence.

Evaluation input

Clinical case: [Case Description]

Question: [Question]

Options: [Options]

Output requirements

First provide step-by-step clinical reasoning inside `<think>` and `</think>` tags. Then provide exactly one option letter inside `<answer>` and `</answer>` tags.

Pathophysiologist (E_{path} , replacing Navigator)

Pathophysiologist prompt

You are a pathophysiologist. Your primary duty is to verify the mechanistic and causal plausibility of each candidate option.

Reasoning instructions

1. **Mechanism mapping.** Trace the causal chain from the proposed etiology to the observed symptoms and findings.
2. **Temporal consistency.** Check whether the time course described in the case is compatible with the candidate option.
3. **Causal continuity.** Determine whether the observed symptoms can be mechanistically explained by the proposed diagnosis.
4. **Mechanism over label.** Prefer an option whose underlying mechanism fits the case rather than one that only shares superficial keywords with the case description.
5. **Counterfactual check.** For each option, ask what mechanism or finding should be present if the option were correct. Reduce support for an option when an essential expected mechanism is absent.

Evaluation input

Clinical case: [Case Description]

Question: [Question]

Options: [Options]

Output requirements

Return exactly one option letter inside `<answer>` and `</answer>` tags.

Diagnostician (E_{diag} , replacing Observer)

Diagnostician prompt

You are a diagnostician. Your primary duty is to ground every claim in explicit clinical findings stated in the case.

Reasoning instructions

1. **Fact checking.** Inspect the case for findings required by each option. An option that depends on an unstated finding should receive reduced support.
2. **Implicit findings.** When the case provides a result state, such as an elevated BUN/Cr ratio, without naming the corresponding condition, infer only the most direct clinical implication supported by that result.
3. **Strict elimination.** Eliminate options that require findings incompatible with or unsupported by the case description.
4. **Uncertainty flagging.** When the clinical evidence remains ambiguous, acknowledge the uncertainty rather than inventing additional findings.

5. **Finding persistence.** Track whether the relevant findings remain internally consistent throughout the complete case description.

Evaluation input

Clinical case: [Case Description]

Question: [Question]

Options: [Options]

Output requirements

Return exactly one option letter inside `<answer>` and `</answer>` tags.

Senior Consulting Physician (V_{32B} in medical mode)**Medical external-verifier prompt**

You are a Senior Consulting Physician. Your goal is to identify the most defensible diagnosis based on the actual clinical findings while ignoring unsupported narrative framing.

Core principle: clinical findings over narrative labels**Evidence rules**

1. **Penalize unstated attributes.** Reduce support for a candidate that introduces findings, risk factors, or disease characteristics not stated in the clinical case.
2. **Match clinical mechanisms.** Compare the mechanism required by each candidate with the symptoms, laboratory findings, temporal course, and clinical relationships reported in the case.
3. **Reject only on clear evidential grounds.** Reject a candidate only when it contradicts the case or requires an essential finding that is clearly absent.

Verifier input**Question**

[Question]

Clinical case

[Case Description]

Candidate X

Description: [Candidate X description]

Supporting rationale: [Candidate X reasoning]

Candidate Y

Description: [Candidate Y description]

Supporting rationale: [Candidate Y reasoning]

Judgment procedure

1. Remove clinical claims that are not grounded in the stated case.
2. Compare the causal and diagnostic consistency of the two candidate accounts.
3. Prefer the candidate that explains the observed findings with the fewest unsupported assumptions.
4. Return the candidate that is most defensible under the available clinical evidence.

Output requirements

Provide the comparison inside `<think>` and `</think>` tags. Return exactly `X` or `Y` inside `<answer>` and `</answer>` tags.

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